Vineeth Raghav Gururajan

Stuttgart, Germany · +49 155 65548480 · vineeth.raghav.g@gmail.com

PROFILE

Working Student, IT Support. M.Sc. Electrical Engineering student at the University of Stuttgart (specialisation Electrical Power Systems) with coursework in IT infrastructure management, networking, operating systems and network security (B.Tech., CGPA 8.70). Enjoys troubleshooting and explaining technical topics to non-technical people (STEM workshop volunteer). Structured, reliable and detail-oriented; interested in process optimisation and AI-based automation (Python, machine learning). English C1. Available up to 20 hours per week, hybrid.

EDUCATION

M.Sc. Electrical Engineering – University of Stuttgart (10/2025 – expected 2027)

- Intelligent Power Grids, Power Systems Engineering, Battery Modeling

B.Tech. Electronics and Computer Engineering – SRM Institute of Science and Technology, Chennai (2021 – 2025), CGPA 8.70/10

- Grades A+ or A in IT Infrastructure Management, Cryptography and Network Security, Embedded Hardware and Operating Systems, Hardware Interfacing and Networking, Database Management Systems, FPGA-Based Embedded Systems, Embedded Hardware and Operating Systems, Microcontrollers and Interfacing, Hardware Interfacing and Networking, IoT System Design, Deep Learning; final-year project graded A+ (10 credits)
- Seminar I (2023, grade O): cryptocurrency price analysis and prediction, review of forecasting methods (technical indicators, linear regression, random forest, LSTM) and evaluation metrics (MAE, RMSE)

PROJECTS

Battery Cell Modeling – team project, University of Stuttgart (2026)

- Analysed cell-to-cell variance caused by electrical, thermal, aging and manufacturing tolerances by simulation; implemented and debugged statistical parameter sampling in the simulation model

Optimized 64-bit RoBA Multiplier with MAC Unit – B.Tech. final-year project, Verilog, Xilinx ISE (2025)

- Designed a rounding-based approximate multiplier with a multiply-accumulate unit that replaces partial-product computation with shift-and-add operations; synthesised in Xilinx ISE (Spartan-3 target) with a reported critical path of 0.575 ns

Duplicate Question Detection – team project (4 students), Python, machine learning (2024)

- Classified duplicate question pairs from the Quora Question Pairs dataset using NLP and fuzzy-ratio features; SVM reached 88.3% and Random Forest 80.02% accuracy; deployed as a web application on Heroku

SKILLS

Electronics and embedded: microcontrollers (ARM, ESP32, Arduino), FPGA design (Verilog, Xilinx ISE), PCB design (Altium), circuit simulation (PSpice), communication protocols (CAN, I2C, SPI, UART)

IT and troubleshooting: debugging and fault finding, networking and network security (coursework), operating systems (coursework), databases (SQL coursework), documentation; Learning goals: Microsoft 365 administration, ticketing systems, device and access management

Automation and programming: Python (machine learning, SVM 88.3% accuracy; web app deployed on Heroku), MATLAB, C/C++, process automation ideas with AI tools

LANGUAGES

English: C1 (IELTS Academic, overall band 7.5, October 2024) · **German:** A2, preparing for the telc B1 exam

CERTIFICATES AND VOLUNTEERING

PSpice Fundamentals (TVS India); Certificate of Training (Livewire Ltd.) · STEM workshops in electronics and coding for underprivileged students; post-cyclone relief work in Chennai